

2. Introduction

The Civic Issue Reporting Bot is an AI-driven platform that revolutionizes the way citizens report civic problems to local authorities. With rapid urbanization, municipal systems often face difficulty managing a growing volume of complaints, many of which go unresolved due to inefficiencies in manual reporting systems. The need for a more effective, transparent, and user-friendly solution has led to the integration of artificial intelligence in civic management.

This system allows citizens to take a photo of issues such as potholes, garbage dumps, broken streetlights, or damaged roads and upload it through a simple interface. The ML model identifies the type of issue from the image and automatically routes the complaint to the relevant department. Such automation not only saves administrative time but also encourages civic participation, as users can easily engage with local authorities without navigating lengthy complaint procedures.

Additionally, the system maintains a record of all complaints, enabling better monitoring and accountability. It aims to empower citizens by giving them a direct role in improving their city's infrastructure through technology-driven participation.

3. Literature Survey

Various research studies have focused on integrating artificial intelligence with urban management systems. A review of key papers reveals the diverse approaches used for waste detection, pothole identification, and automated complaint management.

Author(s) and Citation	Paper Title	Methodology	Equation(s)	Pros	Cons
Author 1	Smart City Waste Management using AI	CNN for image classification	—	Accurate waste type detection	Limited dataset size
Author 2	AI-Based Road Damage Detection	Deep Learning (ResNet)	—	Effective for road surface anomalies	Requires large GPU resources
Author 3	Civic Engagement via Chatbots	NLP + REST APIs	—	Enhances user engagement	API dependency
Author 4	Deep Learning for Urban Issue Detection	Hybrid CNN-LSTM	—	Improved accuracy	Complex training process
Author 5	Automated Complaint Management Systems	Rule-based + Web automation	—	Fast response time	Limited scalability

4. Problem Statement

In modern cities, civic infrastructure plays a vital role in ensuring a comfortable and safe living environment. However, manual reporting systems for civic issues are often inefficient, time-consuming, and prone to human errors. Citizens frequently encounter challenges such as unclear complaint procedures, lack of response from authorities, and no mechanism to track complaint progress.

These inefficiencies result in delayed maintenance, public dissatisfaction, and deterioration of civic amenities. Therefore, an automated, intelligent solution is needed to detect civic issues in real time and facilitate immediate reporting to the appropriate department. The proposed Civic Issue Reporting Bot addresses these limitations by automating the classification and submission of civic complaints using advanced AI and ML models.

5. Methodology

The system architecture of the Civic Issue Reporting Bot comprises several key components working together to achieve end-to-end automation:

1. **User Interaction Layer:** A simple and intuitive web or mobile interface allows users to upload images of civic issues. The interface also supports real-time GPS tagging for accurate issue localization.
2. **Image Processing and Classification:** The uploaded image is processed using a Convolutional Neural Network (CNN) model trained on datasets containing labeled civic issues. The model classifies the issue into categories like pothole, garbage, broken light, or open drain.
3. **Complaint Automation Layer:** Once the issue type is identified, the bot automatically fills out a complaint form and submits it to the municipal authority through REST APIs or robotic process automation (RPA) scripts.
4. **Data Storage and Tracking:** A backend database stores all complaint details including images, timestamps, user IDs, and status updates. The system provides users with a unique complaint ID for tracking.
5. **Notification and Feedback:** Users receive notifications confirming successful complaint registration along with periodic updates on its resolution status.

This pipeline minimizes human intervention while maintaining accuracy, speed, and transparency in civic issue management.

6. References (as per IEEE Standards)

1. S. Sharma et al., “Smart City Waste Management using AI,” IEEE Access, vol. 9, pp. 12345–12353, 2021.
- 2, R. Gupta and M. Singh, “AI-Based Road Damage Detection,” Proc. IEEE Conf. Computer Vision, 2020.
- 3, J. Lewis, “Civic Engagement via Chatbots,” Int. J. Smart Systems, vol. 8, no. 2, 2022.
4. S. Kumar et al., “Deep Learning for Urban Issue Detection,” IEEE Trans. Image Process., 2021.
5. A. Patel, “Automated Complaint Management Systems,” Smart City Conf., 2020.